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SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING

Deep Learning Assignment

Report

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1. Introduction

Deep Neural Networks (DNNs) are a powerful class of artificial neural networks that have demonstrated remarkable success in various machine learning tasks. This report presents an experimental study on the impact of activation functions and network depth on DNN performance. Using the MNIST dataset, we investigate how different activation functions affect gradient propagation and explore the role of depth in enhancing model accuracy.

2. Dataset Used

The MNIST dataset was selected for this study due to its well-structured nature and widespread use in evaluating neural network architectures. It consists of **70,000 grayscale images of handwritten digits (0-9)**, each of size **28x28 pixels**.

- **Training Set:** 60,000 images (normalized between 0 and 1)
- **Test Set:** 10,000 images (normalized between 0 and 1)

The images were flattened into 1D arrays of **784 pixels (28x28)** before being used as input for the neural network models.

3. Methodology

To systematically analyze activation functions and network depth, the following process was followed:

Step 1: Data Preprocessing

- Loaded the MNIST dataset.
- Normalized pixel values to the range $[0, 1]$ to improve model convergence.
- Reshaped the 28x28 images into 1D arrays (784 features) for compatibility with fully connected layers.
- Split the dataset into training and test sets.

Step 2: Implementing Activation Functions

- Built and trained DNN models using **Sigmoid, Tanh, and ReLU** activation functions.
- Evaluated how activation functions impact gradient flow during backpropagation.
- Compared ReLU variants: **ReLU, Leaky ReLU, and Parametric ReLU (PReLU)**.

Step 3: Evaluating Network Depth

- Created models with different numbers of hidden layers (varying depth from 1 to 5).
- Trained each model using the MNIST dataset and recorded validation accuracy and loss.
- Analyzed how deeper networks improve feature extraction but may lead to overfitting.

Step 4: Visualization & Interpretation

- Plotted loss curves to observe gradient behavior.
- Compared test accuracy across different activation functions and network depths.
- Used graphical representations to highlight performance trends.

4. Activation Functions

4.1 Implementing Sigmoid, Tanh, and ReLU

A simple DNN model was implemented using three different activation functions:

- **Sigmoid:** Maps inputs to the range (0,1), useful for binary classification but prone to vanishing gradient problems in deep networks.
- **Tanh:** Maps inputs to the range (-1,1), addressing some of Sigmoid's issues but still suffering from gradient saturation.
- **ReLU (Rectified Linear Unit):** Outputs zero for negative inputs and remains linear for positive inputs, significantly reducing gradient vanishing issues.

4.2 Performance Comparison of ReLU Variants

To further refine ReLU's effectiveness, the following variants were compared:

- **Standard ReLU:** Simple and efficient but suffers from "dying ReLU" neurons.
- **Leaky ReLU:** Allows small negative slopes to avoid inactive neurons.
- **Parametric ReLU (PReLU):** Learns the negative slope dynamically, offering better flexibility.

Key Observations:

- **ReLU outperforms Sigmoid and Tanh**, especially in deep networks, by mitigating vanishing gradients.
- **Leaky ReLU and PReLU** further stabilize training by preventing neurons from becoming inactive.
- **PReLU achieves slightly better accuracy** but increases computational complexity due to trainable parameters.

5. Effect of Network Depth

5.1 Experimenting with Different Depths

To explore the impact of depth, five different architectures were trained, varying the number of hidden layers. The models were assessed based on test accuracy and loss.

Findings:

- **Shallow networks (1-2 layers)** struggled to learn complex patterns.
- **Moderate-depth networks (3-4 layers)** achieved the best balance of accuracy and generalization.
- **Very deep networks (5+ layers)** showed signs of diminishing returns and overfitting.
- **Plots of accuracy vs. depth** revealed an optimal depth threshold where additional layers no longer provided substantial improvements.

6. Visualization of Results

To effectively communicate findings, multiple visualizations were used:

- **Activation Function Comparisons:** Loss curves showed vanishing gradients for Sigmoid and Tanh, while ReLU-based models maintained strong gradients.
- **Depth vs. Accuracy Plot:** Demonstrated the improvement of deeper networks up to an optimal point, beyond which overfitting became apparent.
- **ReLU Variants Comparison:** Highlighted performance differences among standard ReLU, Leaky ReLU, and PReLU.

7. Conclusion

This study explored the impact of activation functions and network depth on DNN performance. Key takeaways include:

- **ReLU is superior** to Sigmoid and Tanh for deep networks due to its resilience against vanishing gradients.
- **Leaky ReLU and PReLU improve upon ReLU** by reducing inactive neurons, with PReLU offering slight additional gains.
- **Depth plays a critical role**, but excessive layers may lead to overfitting and diminishing accuracy improvements.

Through systematic experiments and visualizations, the report highlights best practices for designing deep neural networks with effective activation functions and appropriate depth.